



What Is Registry Analysis?

Registry Analysis means examining the **Windows Registry** to find **system, user, and application activity**.

It is an important part of **digital forensics, malware investigations, and incident response**.

Windows Registry = A **central database** that stores configuration and activity details about:

- Users
- Applications
- System settings
- Connected devices
- Network activity



Registry Structure (Important for Analysis)

Windows Registry has 5 major hives:



HKEY_LOCAL_MACHINE (HKLM)

System-wide settings.

- Installed software
- System drivers
- Hardware details



HKEY_CURRENT_USER (HKCU)

Settings for the logged-in user.

- Recent documents
- Run commands
- User preferences



HKEY_CLASSES_ROOT (HKCR)

File associations (e.g., .pdf → Adobe Reader).



HKEY_USERS (HKU)

Contains all user profiles.



HKEY_CURRENT_CONFIG (HKCC)

Current hardware configuration profile.



What You Can Find During Registry Analysis

◆ 1. Recently Opened Files

HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs

◆ 2. Programs That Auto-Start (Persistence)

HKCU\Software\Microsoft\Windows\CurrentVersion\Run
HKLM\Software\Microsoft\Windows\CurrentVersion\Run
Malware often uses these keys.

◆ 3. USB Device History

SYSTEM\CurrentControlSet\Enum\USBSTOR
Shows pen drives connected to the system.

◆ 4. Installed Applications

HKLM\Software\Microsoft\Windows\CurrentVersion\Uninstall

◆ 5. User Login Information

HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\TypedPaths
Shows run dialog/Explorer search history.

◆ 6. Network Details

Connected Wi-Fi profiles:

HKLM\Software\Microsoft\Windows NT\CurrentVersion\NetworkList\Profiles

Tools Used for Registry Analysis

1. Regedit (Built-in)

Win + R → regedit

View and export registry keys.

2. Registry Explorer (Eric Zimmerman) ★ Best Tool

- Shows deleted keys
- Timestamp details
- Easy-to-read interface

3. Autoruns (Sysinternals)

- Shows startup programs and persistence mechanisms.

4. RECcmd (Eric Zimmerman)

- Command-line registry analysis tool
- Automates extraction of important artefacts.

5. Autopsy / FTK / EnCase

Used in enterprise-level digital forensics.



What You Check in a Real Forensics Case

- ✓ Persistence mechanisms
- ✓ Malware installation traces
- ✓ Account activity
- ✓ File execution history
- ✓ USB usage
- ✓ Browser artifacts
- ✓ Network connection history



Example: Investigating Malware Persistence

You suspect malware, so check:

1. Startup Entries

HKLM\Software\Microsoft\Windows\CurrentVersion\Run

2. Services

HKLM\SYSTEM\CurrentControlSet\Services

3. Scheduled Tasks

HKLM\Software\Microsoft\Windows NT\CurrentVersion\Schedule

If a suspicious entry is found (e.g., svhost.exe), it indicates persistence.